Inputs and source isolation

File: 01_inputs.kicad_sch

Reference low-pass crossover

File: 02_crossover.kicad_sch

Level and phase

File: 03_level_phase.kicad_sch

Subsonic, polarity and output

File: 04_protection_output.kicad_sch

Control and isolated trigger

File: 05_control.kicad_sch

Power and rail supervision

File: 06_power_mute.kicad_sch

Relay drivers and PSU interlock

File: 07_drivers.kicad_sch

HARDWARE RAIL-FAULT INTERLOCK

File: 08_rail_interlock.kicad_sch

REF L/R or LFE -> relay signal/return selection -> buffered mono -> REF LP / AV bypass
-> analogue level -> phase -> HP / AV bypass -> polarity -> output mute -> separate P68

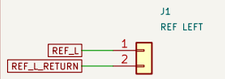
No PCB layout. Separate P68 amplifier unchanged.

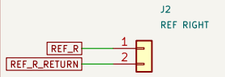
Sheet: /
File: SubController.kicad_sch

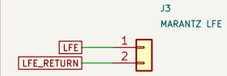
Title: SUBCONTROLLER — ANALOGUE SIGNAL PROCESSING

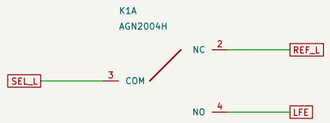
Size: A3	Date: 2026-10-07	**Rev: A — review**
KiCad E.D.A. 10.0.5		Id: 1/9

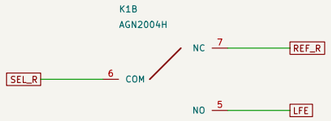
RCA shells MUST be insulated from plate/chassis. No upstream bias or ESD path to board GND.



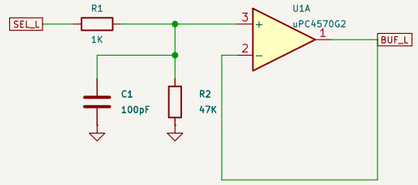


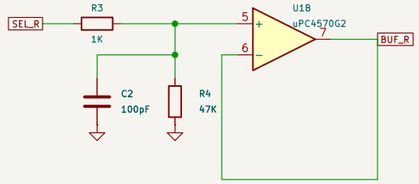




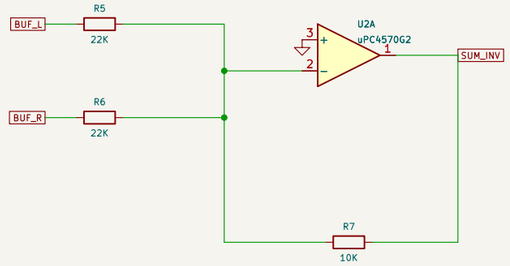


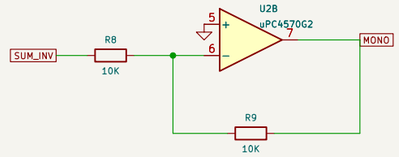
NC = REFERENCE; energised = AV.
K1/K2 switch while K4 muted. Coils on driver sheet.

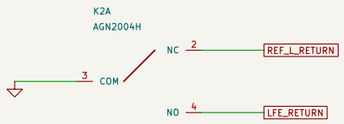


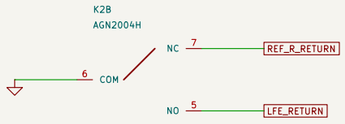


Input loading: 48k incl. series resistor.
Buffer DC gain: 47/48 = 0.979 (−0.18 dB).
100p RF shunt: -1.63 MHz, no bass impact.

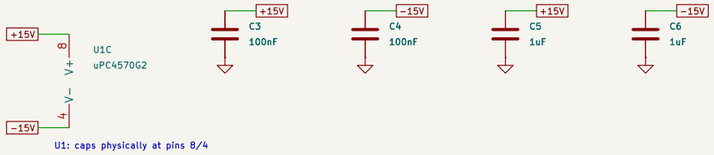




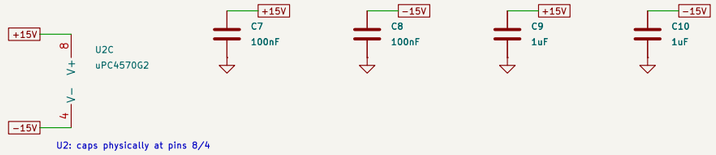




MONO = 0.5L + 0.5R. In AV both inputs receive LFE, so MONO=LFE.
No driver/enclosure boost. Main speakers remain full range.



U1: caps physically at pins 8/4

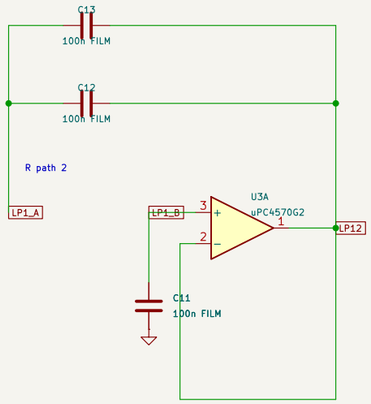


U2: caps physically at pins 8/4

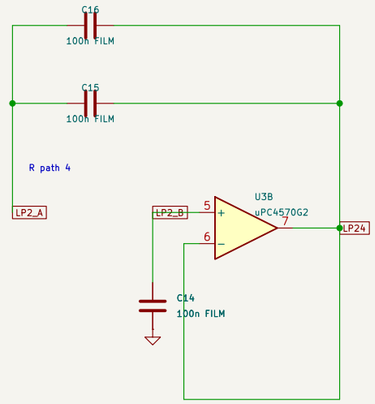
Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Inputs and source isolation/
File: 01_inputs.kicad_sch
Title: Inputs and source isolation

Size: A3	Date: 2026-10-07	**Rev: A — review**
KiCad E.D.A. 10.0.5		Id: 2/9

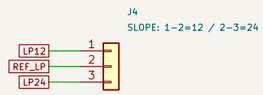
Unity SK: Cf=200n (2x100n), Cg=100n; matched total R1/R2 => Q=0.7071.
RV1: one PCB-mounted PTD904-2015K-B203, four mechanically linked 20k linear gangs.

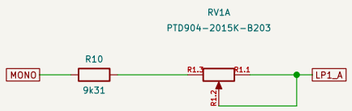


R1/R2 paths continue through RV1 gangs below

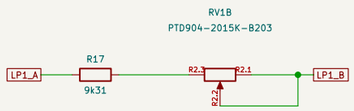


R1/R2 paths continue through RV1 gangs below

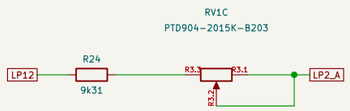




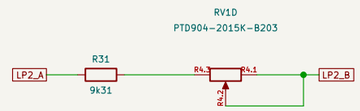
Gang R1: MONO -> LP1_A



Gang R2: LP1_A -> LP1_B

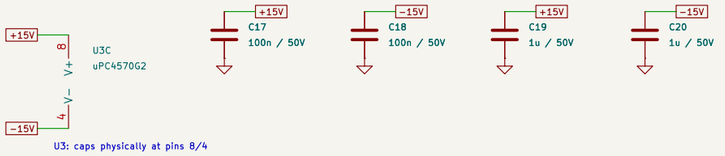


Gang R3: LP12 -> LP2_A



Gang R4: LP2_A -> LP2_B

Nominal R=9k31 + 0..20k: f0=120.9..38.4 Hz. CW increases crossover frequency.
At 70 Hz each rheostat is -6k77. Match gangs and film caps; calibrate actual knob scale.
Wiper tied to unused terminal 1: loss of wiper contact leaves full track resistance.
RV1 body/shaft needs front-edge and plate alignment during future layout: operating ambient <=50 C.



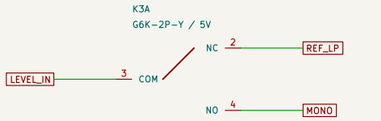
U3: caps physically at pins 8/4

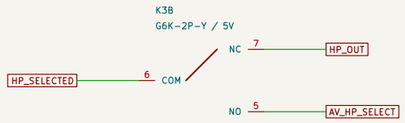
Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Reference low-pass crossover/
File: 02_crossover.kicad_sch

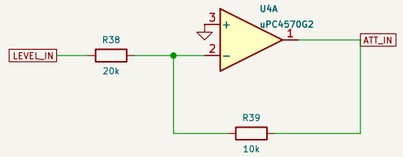
Title: Reference low-pass crossover

Size: A3	Date: 2026-10-07	**Rev: A - review**
KiCad E.D.A. 10.0.5		Id: 3/9

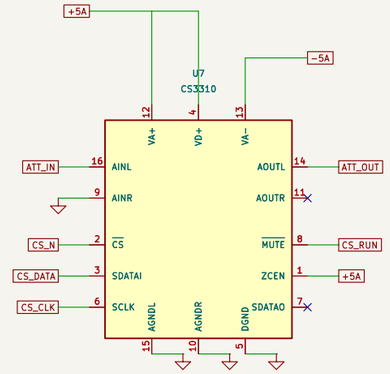
K3 pole A bypasses ONLY the Reference low-pass in AV. Pole B selects AV HP configuration on next sheet.



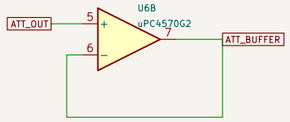




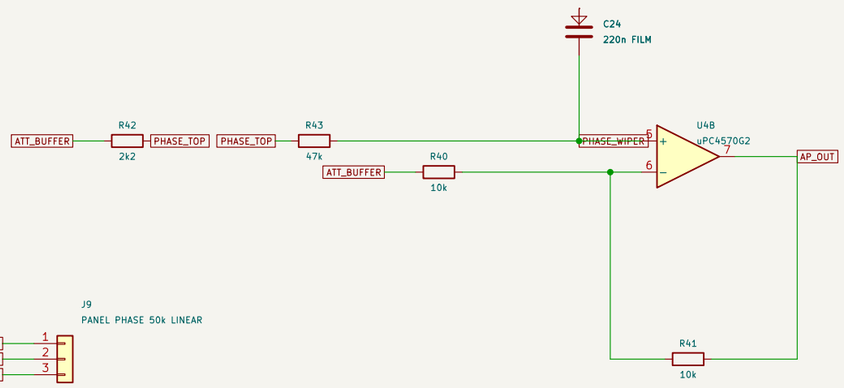
Fixed −0.5 = headroom protection.
Output −2 restores unity overall.

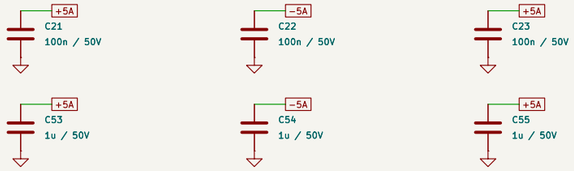


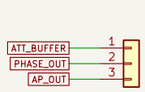
Left channel only. Right input grounded; right gain code=0 (mute).
REF and AV levels stored independently by MCU.
Firmware must prohibit all positive gain codes.

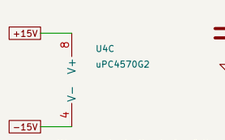


U6B isolates CS output from the HP / all-pass network.
CS sees only op-amp input / short trace capacitance (<100pF).









U4: caps physically at pins 8/4

Panel pot: pin1 top; pin2 wiper; pin3 tied wiper (fail-safe full R).
At 70 Hz: −24 to −137 deg; use bypass for 0 deg. Not constant time delay.

Audio remains analogue; no PCB layout or fabrication approval

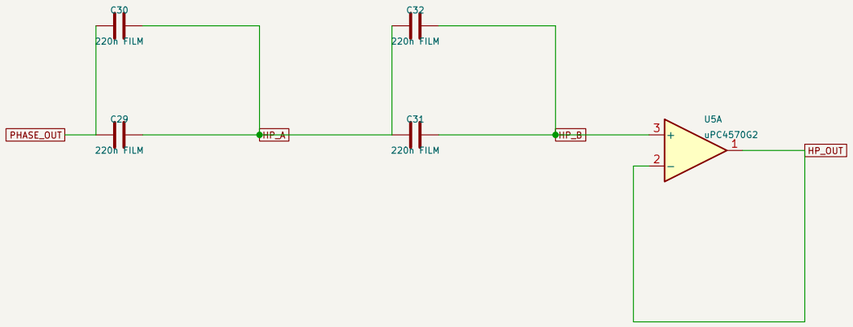
DIY active subwoofer controller

Sheet: /Level and phase/
File: 03_level_phase.kicad_sch

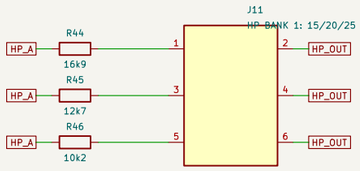
Title: Level and phase

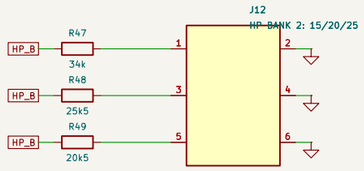
Size: A3	Date: 2026-10-07	**Rev: A — review**
KiCad E.D.A. 10.0.5		Id: 4/9

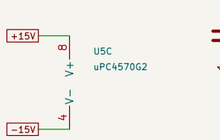
Default REF HP=20 Hz; AV HP bypass preserves AVR response. AV may select HP_OUT when protection required.



R1 bank: HP_A to HP_OUT. R2 bank: HP_B to GND.
C1=C2=440n; Q approx 0.707; unity gain above HP corner.

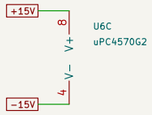




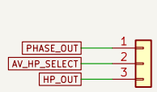


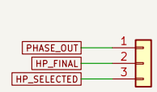
U5: caps physically at pins 8/4

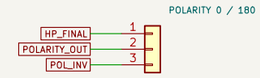


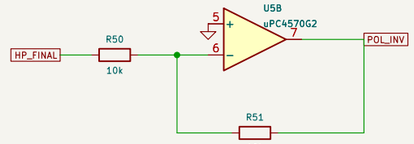


U6: caps physically at pins 8/4

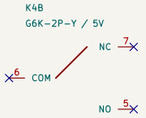


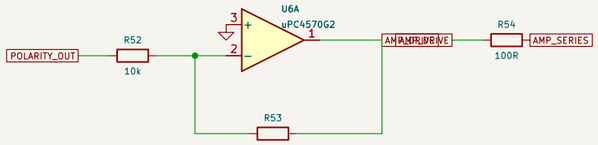


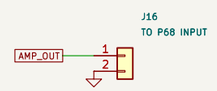


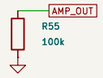












K4 OFF: P68 input grounded; buffer disconnected.
ON: 100R isolates cable capacitance.
P68 nominal 22k / gain 23; confirm actual amplifier.

Audio remains analogue; no PCB layout or fabrication approval

DIY active subwoofer controller

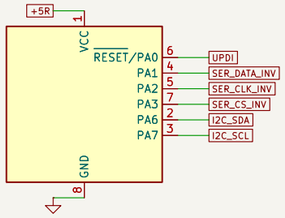
Sheet: /Subsonic, polarity and output/
File: 04_protection_output.kicad_sch

Title: Subsonic, polarity and output

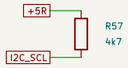
Size: A3	Date: 2026-10-07	**Rev: A — review**
KiCad E.D.A. 10.0.5		Id: 5/9

AUDIO IS NEVER DIGITISED. MCU controls relays and analog attenuation only.
Software GPIO for I2C and inverted 10 kHz serial interface; PA0 reserved UPDI.

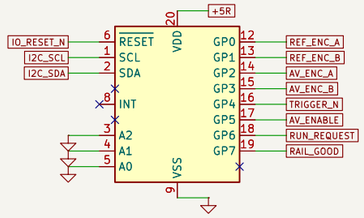
U8
ATtiny412-SS



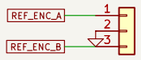




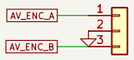
U9
MCP23008-E/SS



J18
REF LEVEL ENCODER

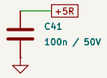


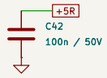
J19
LFE TRIM ENCODER



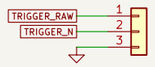






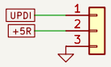


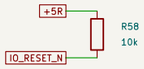
J24
AUTO / REF(open) / AV

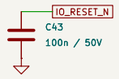


Panel ON–OFF–ON: AUTO / REF / AV. Default shunt 1–2=AUTO.

J17
UPDI / +5R / GND





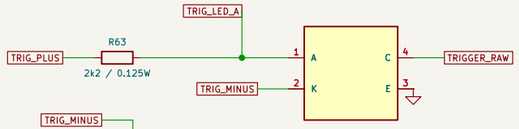


J20
ISOLATED TRIGGER 12V

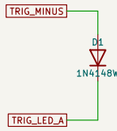


MCP reset: all GPIO inputs; external pulldowns keep relays OFF.
Address pins A2/A1/A0=0 => I2C 0x20.

U10
VO617A-3



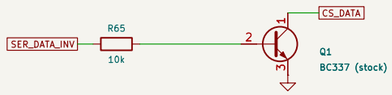


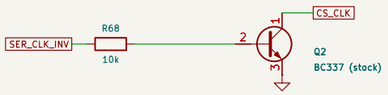




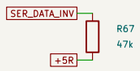


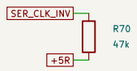


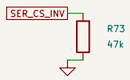












Collector outputs referenced to +5A. No MCU power backfeeds CS3310.
GPIO high => serial low; GPIO low => serial high. Clock/data default low, CS high.

Inactive/open = REF: opto ON = AV. 12V gives ~4.95mA.
For 5V trigger fit 750R instead of 2k2. No trigger return to GND.

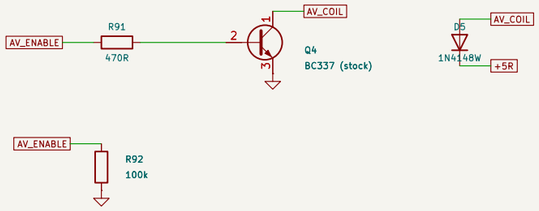
Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Control and isolated trigger/
File: 05_control.kicad_sch
Title: Control and isolated trigger
Size: A3 | Date: 2026-10-07 | **Rev: A – review**
KiCad E.D.A. 10.0.5 | Id: 6/9

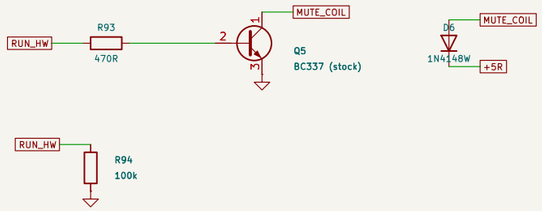
SELV ONLY: externally regulated +/−15V input, NEVER P68 high-voltage power rails.
Nominal +/−15V; maximum +/−15.75V. Supply rating >=250mA positive / >=100mA negative.

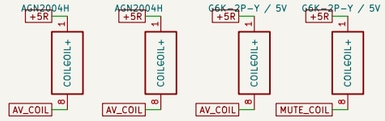
D2
1N4007
D3
1N4007
J21
+15V / GND / −15V
C46
10u / 50V X5R
C44
47u / 25V FR
C45
47u / 25V FR
C57
10u / 50V X5R
C48
10u / 50V X5R
C58
10u / 50V X5R
PWR_FLAG
PWR_FLAG
PWR_FLAG
U11
TPS7A3901
C60
10n C0G / 50V
R105
22k / 0.5%
R106
10k / 0.1%
R107
10k / 0.1%
R111
10k / 0.1%
R108
22k / 0.5%
R109
10k / 0.1%
R110
10k / 0.1%
C59
100n / 50V
C61
10n C0G / 50V
C47
10u / 50V X5R
D4
1N4007
U13
7805 (Diodes stock)
C50
1u / 50V
C51
47u / 25V FR
C49
10u / 50V X5R
U11 −0.27W budget: solder EP to GND copper, thermal vias at layout. U13 +5R stays separate (~1.4W).
C46+C57 and C48+C58: each pair directly at IN pins. C47/C49 directly at OUT pins.
NR/SS100n: CFF10n each. Keep CS VD+ tied to VA+. Existing rail-fault/output mute retained.
TPS feedback: + arm (22k+10k)/10k -> +4.990V; − arm (22k+10k+10k)/10k -> −4.998V.
Negative lower arm returns to BUF (pin8), NOT GND. EN tied to +15V; EP11 tied to GND.
R74
680R
U14
LM4040A25IDBZR
R75
1k
R76
9k09
R77
15k
R78
10k
R79
6k2
R80
93k1
R81
18k / 0.1%
R82
7k15 / 0.1%
R83
100k
R84
20k
R85
10k
R86
100k
R87
10k / 0.1%
R88
10k / 0.1%
R89
10k / 0.1%
R90
30k / 0.1%
U15E
LM339
C52
100n / 50V
U15A
LM339
U15B
LM339
U15C
LM339
U15D
LM339
Trip approx +13.5V / −14V / +4.69V / −4.66V.
Open-collector RAIL_GOOD is read by MCU; hardware also overrides RUN.
No positive feedback; firmware rearm delay prevents repeated unmute.

Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Power and rail supervision/
File: 06_power_mute.kicad_sch
Title: Power and rail supervision
Size: A3 | Date: 2026-10-07 | **Rev: A — review**
KiCad E.D.A. 10.0.5 | Id: 7/9

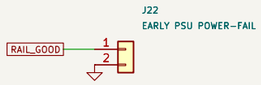
Default: AV_ENABLE=0, RUN_REQUEST=0. Output stays muted until firmware sequence completed.
Rail interlock clamps RUN_HW independently; MCU requires fresh settling delay after fault.



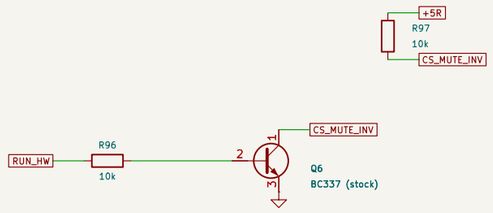


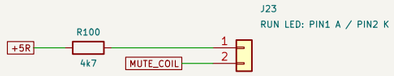


Q4 switches three 5V relay coils (~60mA).
Q5 switches K4 coil (~20mA). Flyback diode slows release:
wait >=30ms after mode-relay operation.
RUN_HW ~1.7-2.1V ON (base-drive bus, not a CMOS output).



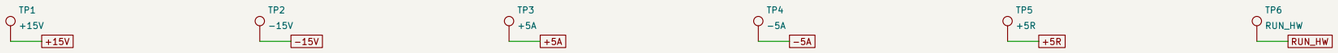






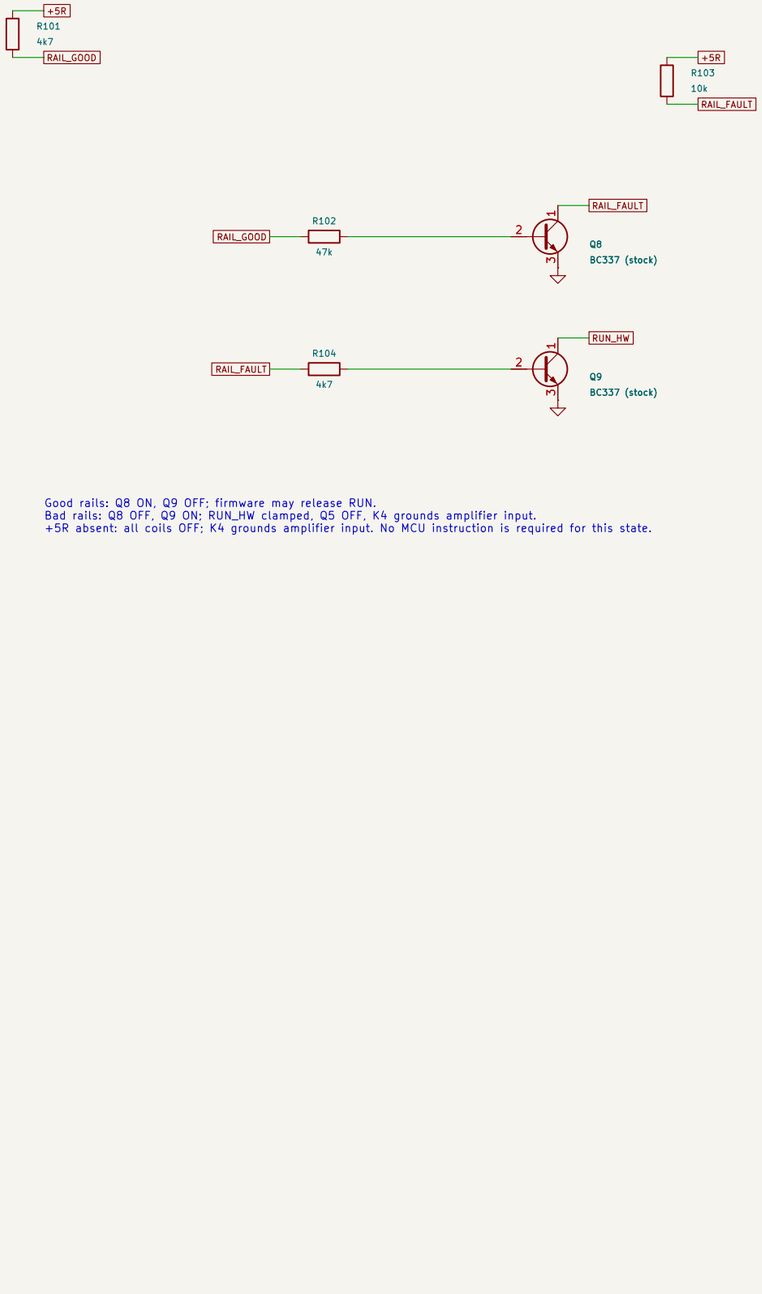
Optional open collector contact to GND. Assert before analog rails collapse.
NO external voltage injection. Without early warning, arbitrary fast
power loss cannot be guaranteed pop-free by a mechanical relay.

RUN=0: Q6 OFF, Q7 ON => CS MUTE asserted.
RUN=1: Q6 ON, Q7 OFF => CS_RUN=+5A.
Collector interface prevents +5R backfeeding +5A.



Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Relay drivers and PSU interlock/
File: 07_drivers.kicad_sch
Title: Relay drivers and PSU interlock
Size: A3 | Date: 2026-10-07 | **Rev: A — review**
KiCad E.D.A. 10.0.5 | Id: 8/9

Rail comparators / external early power fail pull RAIL_GOOD low. GP7 reads it independently of RUN_REQUEST.
On a fault: mute immediately in hardware; firmware clears RUN_REQUEST and waits 500ms stable RAIL_GOOD before rearm.



Good rails: Q8 ON, Q9 OFF; firmware may release RUN.
Bad rails: Q8 OFF, Q9 ON; RUN_HW clamped, Q5 OFF, K4 grounds amplifier input.
+5R absent: all coils OFF; K4 grounds amplifier input. No MCU instruction is required for this state.

Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller

Sheet: /HARDWARE RAIL–FAULT INTERLOCK/
File: 08_rail_interlock.kicad_sch

Title: HARDWARE RAIL–FAULT INTERLOCK

Size: A3	Date: 2026–10–07	**Rev: A – review**
KiCad E.D.A. 10.0.5		Id: 9/9